FUNCTIONAL AND NON FUNCTIONAL REQUIREMENTS

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale (short)
FR-001	Functional	The system shall allow faculty to record attendance for students for a selected subject.	High	When attendance is submitted, the record is saved and appears in the student's attendance details.	Required for maintaining attendance records.
FR-002	Functional	The system shall calculate the attendance percentage for each student and subject.	Medium	For valid attendance data, the displayed percentage matches the calculated value based on classes attended and total classes.	Helps students track attendance
FR-003	Functional	The system shall allow faculty to enter and update marks for each student.	high	After marks are submitted, the updated marks are displayed in the student's academic record	Required for maintaining academic performance.
FR-004	Functional	Calculate the semester GPA using grade points derived from student marks and the corresponding subject credits, according to the institution's prescribed formula.	High	When valid grades and credits are available, the calculated GPA matches the institution's grading formula.	Provides academic performance information

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FR-005	Functional	The system shall allow students to view their attendance, marks, grades, GPA, and CGPA.	Medium	After successful login, the student can access all available academic records associated with their account.	Provides students access to their records.
FR-006	Functional	The system shall allow users to log in and log out securely	High	Valid credentials allow access; invalid credentials are rejected.	Controls access to the system.
FR-007	Functional	The system shall allow authorized faculty to view student profile details.	Medium	Authorized faculty can retrieve the correct student's profile.	Helps faculty identify students.
FR-008	Functional	The system shall maintain subject details and corresponding credit values.	High	Each subject has a valid credit value stored in the system	Supports GPA calculations.

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FR-009	Functional	Calculate CGPA using semester results and the corresponding subject credits, according to the institution's prescribed formula.	High	CGPA matches the institution's prescribed calculation formula.	Maintains cumulative academic results
FR-010	Functional	Convert student marks into grade points according to the institution's grading rules.	High	Marks are converted using the configured grading rules.	Ensures consistent grade-point calculations.
FR-011	Functional	Allow students to select a semester and view the academic records associated with that semester.	Medium	The student can select a semester and view its academic records.	Helps students track progress.
FR-012	Functional	The system shall allow faculty to search for a student using the student's ID.	Medium	Searching with a valid student ID returns the matching student.	Speeds up record retrieval.
FR-013	Functional	The system shall allow faculty to filter attendance records by subject and date	Medium	Selected filters return only matching attendance records.	Makes attendance review easier.

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FR-014	Functional	The system shall allow authorized faculty to update previously recorded attendance.	High	Updated attendance is saved and reflected in the student's records.	Corrects attendance entry errors
FR-015	Functional	The system shall validate entered marks against the maximum permitted marks.	High	Marks within the permitted range are accepted; out-of-range marks are rejected.	Prevents invalid marks.
FR-016	Functional	The system shall generate a class-wise attendance report for faculty.	Medium	The report displays attendance details for students in the selected class or subject.	Helps faculty monitor attendance
FR-017	Functional	Notify students when their attendance percentage falls below the minimum attendance threshold specified by the institution.	Medium	A notification is generated when attendance falls below the configured threshold.	Alerts students to low attendance
FR-018	Functional	The system shall allow faculty to view a subject-wise summary of students' marks.	Medium	Faculty can see the marks of students enrolled in the selected subject.	Supports class performance analysis.

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FR-019	Functional	The system shall allow students to download their academic records as a report	Medium	The downloaded report contains the student's available academic records.	Provides a portable academic summary.
NFR001	Nonfunc.	The system shall display requested attendance and academic records within 2 seconds for at least 95% of requests under normal operating conditions.	Medium	At least 95% of requests display the requested records within 2 seconds under normal operating conditions	Ensures quick access to records.
NFR002	Nonfunc.	The system shall restrict access to academic records based on user roles. Students shall access only their own academic records, while faculty shall access records they are authorized to view or manage.	High	An unauthorized user attempting to access another student's record is denied access in every tested attempt.	Protects confidential student information.
NFR003	Nonfunc.	The system shall provide a user-friendly interface for students and faculty. At least 4 out of 5 test users shall be able to access attendance and GPA records without assistance.	Medium	At least 4 out of 5 test users can access attendance and GPA records without assistance	Makes the system easy to use.

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NFR004	Nonfunc	The system shall be available for at least 99% of the scheduled college operating hours.	High	The system shall be accessible for at least 99% of the scheduled operating hours during testing.	Ensures users can access records when required.
NFR005	Nonfunc	The system shall support backup and recovery of student records and allow a deleted test record to be successfully restored from a backup	Medium	A backup can be created and a deleted test record can be successfully restored from the backup	Prevents permanent loss of student data.
NFR006	Nonfunc	The system shall maintain the accuracy and consistency of student academic records and prevent unauthorized or invalid modifications.	High	Valid updates are saved correctly, and invalid modifications are rejected without corrupting existing records.	Ensures the reliability and accuracy of academic records.